

An-Najah National University

Faculty of Engineering and Information Technology

Data Warehousing Assignment

Movie Rental Data Warehouse Design

Student Name: Tala Shilleh

Student ID: 12240960

Course: Database 2

Instructor: Moamen Abu Ghazaleh

Semester: Second Semester 2025/2026

Contents

1	Introduction	2
2	Business Questions	2
3	Dimensional Model Design	2
3.1	Selected Business Processes	2
3.2	Fact Tables	3
3.2.1	Fact_Rental	3
3.2.2	Fact_Payment	3
3.3	Dimension Tables	4
3.3.1	Dim_Date	4
3.3.2	Dim_Customer	4
3.3.3	Dim_Film	5
3.3.4	Dim_Store	5
3.3.5	Dim_Staff	6
4	Dimensional Model Diagram	6
5	ETL Design	9
5.1	Extract	9
5.2	Transform	9
5.3	Load	10
6	Data Quality Considerations	11
7	Analytical Reports and Visualizations	12
7.1	Top Customers by Spending	12
7.2	Top Rented Movies	12
7.3	General Statistics	13
8	Conclusion	14

1 Introduction

This assignment presents a dimensional Data Warehouse design for a movie rental business using the Sakila OLTP database.

The original OLTP system is designed for daily operational transactions such as rentals, payments, customer management, and inventory operations. Although the OLTP system is efficient for transaction processing, it is not optimized for analytical reporting because the data is highly normalized and distributed across many tables.

The purpose of the proposed Data Warehouse is to reorganize operational data into a dimensional model that supports business intelligence and analytical reporting.

The proposed solution uses a Star Schema design with fact tables and shared dimension tables to simplify reporting queries and improve analytical performance.

The Data Warehouse enables management to analyze customer spending, movie popularity, rental activity, and store performance more efficiently.

2 Business Questions

The proposed Data Warehouse is designed to answer important analytical business questions related to the movie rental business.

1. Which films are rented most frequently?
2. Which customers generate the highest revenue?
3. Which stores generate the highest number of rentals?
4. Which staff members process the highest number of rentals or payments?
5. How does rental activity change over time?
6. Which movie categories are most popular?
7. What are the total revenues generated from rentals?

These analytical questions help management understand customer behavior, identify profitable films, evaluate operational performance, and support strategic business decision-making.

3 Dimensional Model Design

3.1 Selected Business Processes

The selected business processes from the OLTP system are:

- Rental transactions
- Payment transactions

These processes represent the core operational activities that require analytical reporting.

The proposed dimensional model uses a Star Schema because it simplifies querying and improves reporting performance.

3.2 Fact Tables

3.2.1 Fact_Rental

Business Process:

Rental transaction analysis.

Grain:

One row for each rental transaction.

Measures:

- rental_count

Related Dimensions:

- Dim_Date
- Dim_Customer
- Dim_Film
- Dim_Store
- Dim_Staff

3.2.2 Fact_Payment

Business Process:

Payment and revenue analysis.

Grain:

One row for each payment transaction.

Measures:

- payment_amount
- payment_count

Related Dimensions:

- Dim_Date
- Dim_Customer
- Dim_Store
- Dim_Staff

3.3 Dimension Tables

3.3.1 Dim_Date

Main Attributes:

- date_key
- full_date
- day_number
- month_number
- month_name
- quarter_number
- year_number

Source OLTP Tables:

Rental and payment date fields.

Purpose:

Used for time-based analysis and trend reporting.

3.3.2 Dim_Customer

Main Attributes:

- customer_key
- customer_id
- full_name
- email
- city

- country

Source OLTP Tables:

customer, address, city, country

Purpose:

Used to analyze customer activity and revenue generation.

3.3.3 Dim_Film

Main Attributes:

- film_key
- film_id
- title
- category_name
- language_name
- rental_rate
- rental_duration
- release_year

Source OLTP Tables:

film, category, film_category, language

Purpose:

Used to analyze movie popularity and rental performance.

3.3.4 Dim_Store

Main Attributes:

- store_key
- store_id
- city
- country
- manager_name

Source OLTP Tables:

store, address, city, country, staff

Purpose:

Used to analyze store activity and compare branch performance.

3.3.5 Dim_Staff

Main Attributes:

- staff_key
- staff_id
- full_name
- store_id

Source OLTP Tables:

staff

Purpose:

Used to analyze staff activity and operational performance.

4 Dimensional Model Diagram

The following diagram shows the Star Schema design for the Movie Rental Data Warehouse.

- Dim.Film
- Dim.Store
- Dim.Staff

The main fact tables are:

- Fact.Rental
- Fact.Payment

The Star Schema design simplifies analytical queries and improves reporting performance.

5 ETL Design

5.1 Extract

The extraction process collects data from the Sakila OLTP database tables required for building the dimensional model.

The extracted tables include:

- rental
- payment
- customer
- film
- inventory
- store
- staff
- address
- city
- country
- category
- film_category
- language

These operational tables provide the raw data needed to build the Data Warehouse dimensions and fact tables.

5.2 Transform

Several transformations are applied during the ETL process before loading the warehouse.

- Related OLTP tables are joined to build complete dimensions.
- Customer and staff full names are generated by combining first and last names.
- Date keys are generated from rental and payment dates.
- Surrogate keys are automatically generated for dimensions.

- Missing and duplicate records are cleaned.
- Country and city names are standardized.
- Payment records are transformed into analytical fact tables.
- Rental records are transformed into analytical fact tables.

These transformations prepare the data for business intelligence and analytical reporting.

5.3 Load

The loading process inserts transformed data into the Data Warehouse.

Dimension tables are loaded first because fact tables depend on dimension surrogate keys.

The loading order is:

1. Dim_Date
2. Dim_Customer
3. Dim_Film
4. Dim_Store
5. Dim_Staff
6. Fact_Rental
7. Fact_Payment

Fact tables reference surrogate keys from dimension tables to maintain relationships between business events and descriptive dimensions.

Data validation checks are applied during loading to ensure data consistency and accuracy.

6 Data Quality Considerations

Several data quality checks are applied during the ETL process to ensure reliable analytical reporting.

- Missing values are detected and handled before loading.
- Duplicate customer, rental, and payment records are removed.
- Foreign key relationships between dimensions and facts are validated.
- Date fields are standardized using consistent formats.
- Customer names and city names are standardized.
- Invalid records are excluded from loading.
- Payment values are validated to ensure positive amounts.
- Null values in dimension mappings are checked before loading fact tables.

These data quality processes improve consistency, accuracy, and reliability of analytical reports.

7 Analytical Reports and Visualizations

The following charts present analytical insights generated from the Data Warehouse after executing ETL processes and analytical SQL queries.

These visualizations help management understand customer spending behavior and movie rental trends.

7.1 Top Customers by Spending

The following chart shows the customers with the highest total spending.

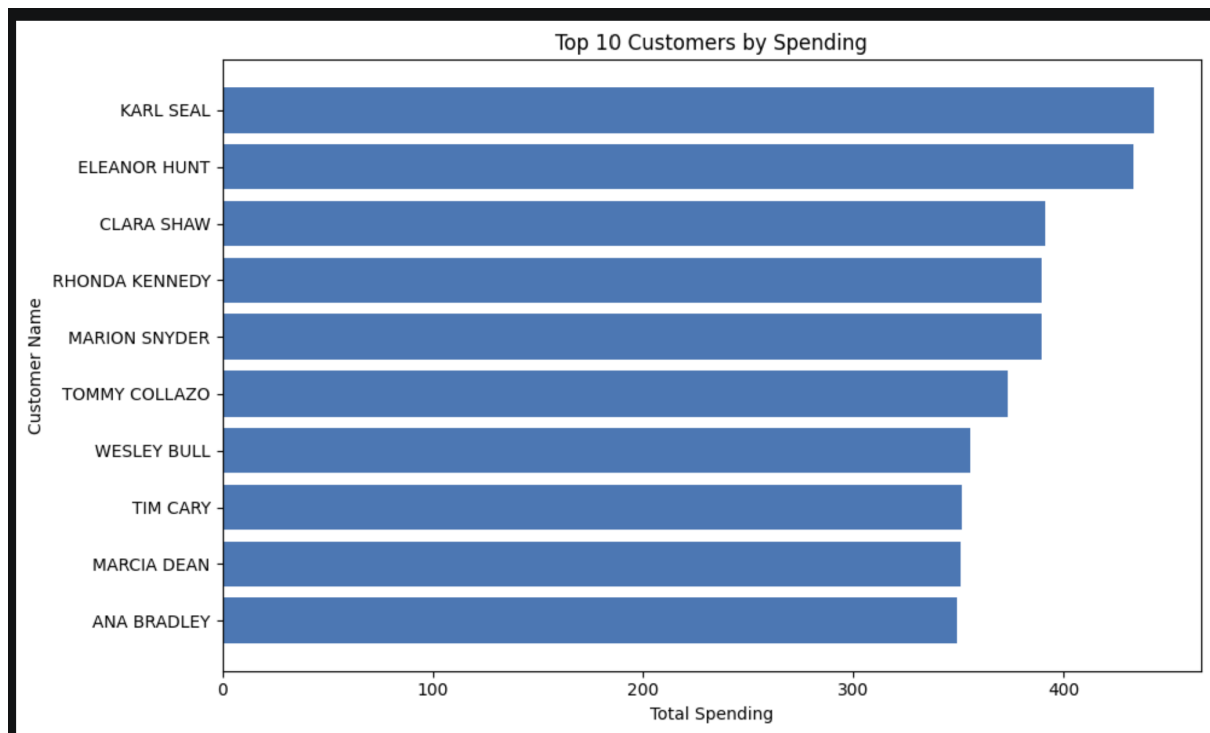


Figure 2: Top Customers by Spending

This analysis helps management identify high-value customers and support customer relationship management strategies.

7.2 Top Rented Movies

The following chart presents the most rented movies in the system.

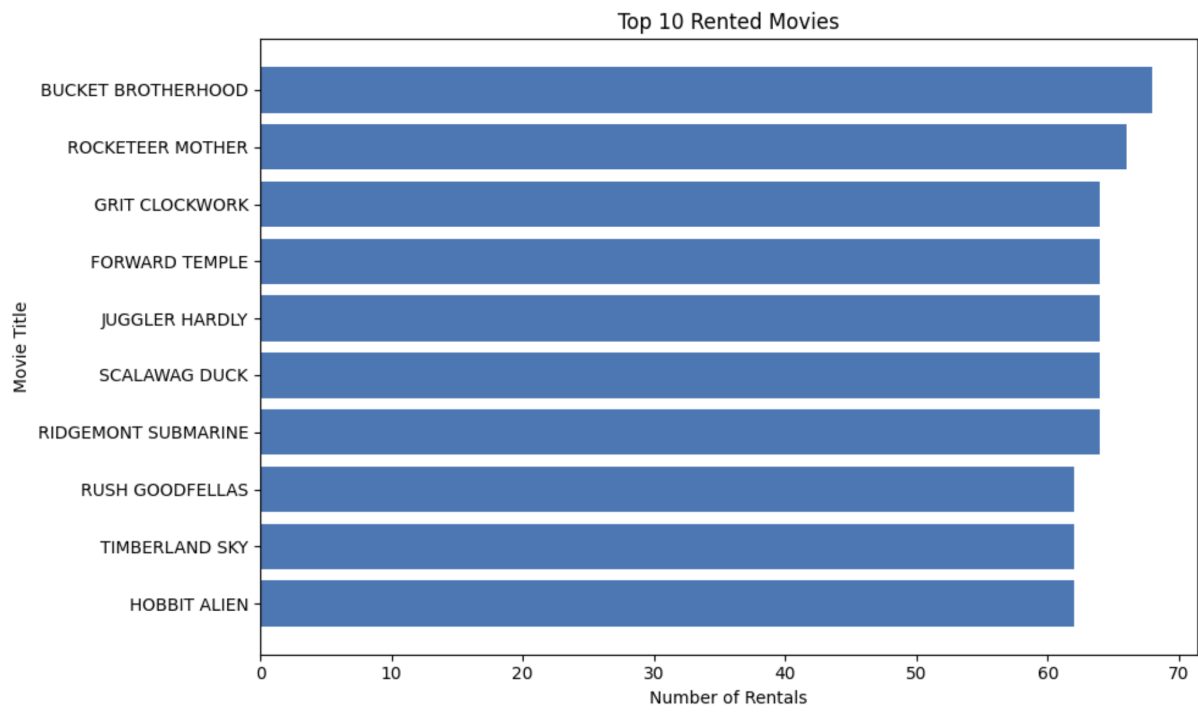


Figure 3: Top Rented Movies

This analysis helps identify popular movies and supports inventory and marketing decisions.

7.3 General Statistics

The Data Warehouse also provides several business statistics including:

- Total Customers
- Total Films
- Total Stores
- Total Rentals
- Total Payments
- Total Revenue

These statistics provide a high-level overview of the movie rental business operations.

8 Conclusion

This assignment presented a dimensional Data Warehouse design for the Sakila movie rental system.

The proposed solution reorganizes transactional operational data into a Star Schema model containing fact tables and dimension tables that support analytical reporting and business intelligence.

The ETL process extracts operational data from the OLTP system, transforms and cleans the data, and loads it into the Data Warehouse in a structured analytical format.

The implemented Data Warehouse enables management to analyze customer spending, rental activity, movie popularity, store performance, and business trends more efficiently.

Using a Star Schema design improves reporting performance, simplifies analytical queries, and supports better business decision-making.

Overall, the project provides a scalable foundation for future analytical processing and business intelligence solutions in the movie rental business.